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Original Article

Species Differences in Developmental Toxicity of Epoxiconazole and Its Relevance to Humans

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Epoxiconazole, a triazole-based fungicide, was tested in toxicokinetic, prenatal and pre-postnatal toxicity studies in guinea pigs, following oral (gavage) administration at several dose levels (high dose: 90 mg/kg body weight per day). Maternal toxicity was evidenced by slightly increased abortion rates and by histopathological changes in adrenal glands, suggesting maternal stress. No compound-related increase in the incidence of malformations or variations was observed in the prenatal study. In the pre-postnatal study, epoxiconazole did not adversely affect gestation length, parturition, or postnatal growth and development. Administration of epoxiconazole did not alter circulating estradiol levels. Histopathological examination of the placentas did not reveal compound-related effects. The results in guinea pigs are strikingly different to those observed in pregnant rats, in which maternal estrogen depletion, pathological alteration of placentas, increased gestation length, late fetal death, and dystocia were observed after administration of epoxiconazole. In the studies reported here, analysis of maternal plasma concentrations and metabolism after administration of radiolabeled epoxiconazole demonstrated that the different results in rats and guinea pigs were not due to different exposures of the animals. A comprehensive comparison of hormonal regulation of pregnancy and birth in murid rodents and primates indicates that the effects on pregnancy and parturition observed in rats are not applicable to humans. In contrast, the pregnant guinea pig shares many similarities to pregnant humans regarding hormonal regulation and is therefore considered to be a suitable species for extrapolation of related effects to humans. *Birth Defects Res (Part B)* 98:230–246, 2013. © 2013 Wiley Periodicals, Inc.

Key words: ***Epoxiconazole; prenatal toxicity; postnatal toxicity; species differences; guinea pig; rat; developmental toxicity***

INTRODUCTION

Epoxiconazole [chemical name: (2RS,3SR)-1-[3-(2- chlorophenyl)-2,3-epoxy-2-(4-fluorophenyl)-propyl]-1H-1,2,4-triazole CAS-No. 133855–98–8] is used as an effective fungicide in agriculture. As with other triazole-derived fungicides, antifungal activity is mediated through inhibition of ergosterol biosynthesis. Inhibition of lanosterol-C14-demethylase (CYP51) leads to accumulation of methylated sterols in the fungal membrane, thereby impairing its function. However, enzyme inhibition is not restricted to CYP51 (a fungal enzyme). Other enzymes inhibited by triazole fungicides comprise aromatase (CYP19), which converts androstenedione to estrogen, and 17α-hydroxylase (CYP17), which converts progesterone into 17α-hydroxy-progesterone, a precursor of cortisol and androsterone (Goetz et al., 2007). 17α-Hydroxylase also converts pregnenolone into 17α-pregnenolone, a precursor of dihydroepiandrosterone, which is further metabolized to estrone and estradiol by aromatase. Thus, aromatase and 17α-hydroxylase are key enzymes in steroidogenesis. Inhibition of CYP26 isoenzymes, which are involved in the metabolism of the morphogen retinoic acid, has also been described for triazoles and is discussed as a probable mechanism for the teratogenesis observed with these compounds (Menegola et al., 2006).

Taxvig et al. (2007) administered epoxiconazole orally at dose levels of 15 or 50 mg/kg body weight per day to pregnant rats from gestation day 7 (GD 7) to either GD 21 or to postnatal day 16. Maternal animals had increased

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gestational length, dystocia, and an increased incidence of late intrauterine resorptions and stillbirths at 50 mg/kg body weight per day. Decreased pup weights and increased postnatal deaths were also observed. No toxicologically relevant effects were observed at 15 mg/kg body weight per day. Increased incidences of late intrauterine resorptions were confirmed in a subsequent study after oral administration of 50 mg/kg body weight per day (Stinchcombe et al., 2013). However, the dose of 50 mg/kg body weight per day also elicited distinct maternal toxicity, as shown by reduced corrected body weight, reduced food consumption, clear anemia, and almost complete suppression of maternal serum estradiol. Placentas of fetuses undergoing late resorption showed degenerative changes of the labyrinth and trophospongium (Rey Moreno et al., 2013). These results indicate that intrauterine resorptions are not caused by a specific embryofetotoxic effect of epoxiconazole, but rather due to a maternally mediated effect at toxic dose levels on the placentas, which are no longer able to adequately nourish the fetuses. Both the increased incidence of late resorptions and degeneration of the placentas could be prevented by maternal supplementation with estrogen (Rey Moreno et al., 2013; Stinchcombe et al., 2013). However, this supplementation did not prevent the other maternally toxic effects of epoxiconazole, that is, reduced body weight gain and anemia. Therefore, it is evident that estradiol homeostasis plays a critical role in epoxiconazole mediated late intrauterine resorptions, and that these effects are due to aromatase inhibition in rats.

The rat, as a model for effects mediated by aromatase inihibition, has been criticized as a species that is not relevant for humans (Scientific Committee on Plants, 1999); instead, the guinea pig was recommended as a suitable species for assessing human risk. More recently, the applicability of the guinea pig for regulatory reproductive toxicology testing has also been acknowledged (Rocca and Wehner, 2009).

Few publications from the last decades reviewed the characteristics of guinea pig reproductive biology in comparison to a variety of species including humans. There is general consensus that the guinea pig, which displays several key physiological characteristics of gestation that more closely resemble human pregnancy than do currently established standard animal models, is a valid alternative to murid rodents. Examples of these physiological characteristics comprise hormonal regulation (Batra et al., 1980; Hobkirk and Glasier, 1993), placentation (Carter, 2007), and parturition (Mitchell and Taggart, 2009). As in humans, the guinea pig placenta is hemomonochorial (Enders, 1965; Carter and Enders, 2004; Carter, 2007) and the trophoblast invades extensively into the endometrium.

Therefore, the guinea pig was used in the present studies as an alternative animal model for developmental and reproductive toxicity. These studies were performed to examine whether the effects of epoxiconazole on the hormonal regulation of pregnancy, placenta morphology, embryonic/fetal development, and parturition as previously described in rats were species-specific.

Three studies were performed in guinea pigs to address endpoints during gestation and early postnatal development:

(1) A toxicokinetic study in pregnant guinea pigs to establish plasma levels and metabolite profiles after oral administration of radiolabeled epoxiconazole.
(2) A prenatal toxicity study with endpoints measuring maternal and fetotoxicity terminated by cesarean section.
(3) A pre-postnatal toxicity study with endpoints examining maternal toxicity, gestation length, parturition, and postnatal development until weaning.

MATERIAL AND METHODS

General

All studies except analyses of hormone levels were conducted according to the OECD Principles of Good Laboratory Practice (OECD, 1992), which principally meet the United States Environmental Protection Agency Good Laboratory Practice Standards [40 CFR Part 160 (FIFRA) and Part 792 (TSCA)].

The toxicokinetic study was performed according to OECD Guideline 417 (OECD, 2010). The design of the modified prenatal study followed OECD Guideline 414 and EPA OPPTS 870.3700 (US EPA, 1998; OECD, 2001), the pre- and postnatal toxicity study was performed referring to ICH S5A, Section 4.1.2 (ICH, 2005), but terminated 3 weeks after weaning. In the latter two studies, the treatment periods were adapted to the longer gestation period of the guinea pig, which takes approximately 69 days rather than 21 days in rats.

Test Compound

Epoxiconazole ((2RS,3SR)-1-[3-(2-chlorophenyl)-2,3-epoxy-2-(4-fluorophenyl)-propyl]-1H-1,2,4-triazole), batch COD-001118, was obtained from BASF (Ludwigshafen, Germany) and had a purity of >97%.

In addition to the unlabeled batch, ^{14}C-labeled epoxiconazole with a radiochemical purity of 99.9% was used in the toxicokinetic study. It was mixed with the nonlabeled substance to yield a specific activity of about 1 MBq/mg (5 and 15 mg/kg body weight per day) or 0.5 MBeq/mg (50 and 90 mg/kg body weight per day), respectively.

Dosing formulations of epoxiconazole were prepared with 1% carboxymethylcellulose suspension in highly deionized water.

Test Animals

The studies were performed in an AAALAC-approved laboratory in accordance with the German Animal Welfare Act and the effective European Council Directive. Dunkin-Hartley guinea pigs (Crl:HA) were supplied by Charles River Laboratories, Research Models and Services, Germany GmbH. Only animals free from clinical signs of disease were used for the investigations.

In the toxicokinetic study and the prenatal toxicity study, the breeder paired female guinea pigs on the day after they had given birth to a litter, using the postpartal estrus to obtain time-mated animals. Thus, no nulliparous females were used in these experiments. In the pre-postnatal toxicity study BASF Experimental Toxicology and Ecology Laboratories mated female guinea pigs in the estrous phase with a male partner for several hours (morning or afternoon pairing) until copulation was observed.

Guinea pigs were housed individually on dust-free wooden bedding in type H-Temp (PSU) cages supplied by Tecniplast, Germany (floor area about 2065 cm^2), in fully air-conditioned rooms with a range of temperature of 20–24°C and a range of relative humidity of 30–70%. The air exchange rate was 15 times per hour. The day/night cycle was generally 12 hr (12 hr light from 6:00 AM to 18:00 PM and 12 hr darkness from 18:00 PM to 6:00 AM). Food (pelleted Kliba maintenance diet rabbits and guinea pigs, "GLP" meal, supplied by Provimi Kliba SA, Kaiseraugst, Switzerland) and drinking water (supplied from water bottles) was available to the animals *ad libitum* throughout the study.

Experimental Procedure

Toxicokinetic study. Groups of six pregnant guinea pigs received unlabeled epoxiconazole orally via gavage daily from GD 6 to GD 9 at dose levels of 5, 15, 50, or 90 mg/kg body weight, followed by administration of radiolabeled substance on GD 10. The dose volume was 10 mL/kg body weight, the animals were not fasted before treatment. Blood samples (approximately 100 μL) were taken from the ear vein (at 0.5, 1, 2, 4, 8, 24, 48, and 72 hr) or by exsanguination under isoflurane anesthesia (96 hrs) after administration of radiolabeled epoxiconazole.

The total radioactivity in plasma was determined using liquid scintillation counting after mixing the samples with appropriate amounts of Hionic Fluor cocktail. Analysis of the kinetic data (C_{max}, AUC, half-life) was performed using the PC program system TOPFIT, version 2.0. Only pregnant animals were used for evaluation.

For metabolite profiling, pooled plasma samples were prepared by combining equal volumes from each animal of a certain dose group for the respective time periods, and the total radioactive residues in the pooled samples were measured. Proteins were precipitated from the pooled plasma samples with acidified acetonitrile, and the protein precipitates were incubated with proteases. The supernatants after protein precipitation and the solubilized residues after protease treatment were analyzed by LSC and HPLC. Separation of peaks was performed via gradient elution on reversed-phase columns or on a HILIC column.

Prenatal toxicity study. Groups of 30 mated guinea pigs received epoxiconazole orally via gavage once daily from implantation to sacrifice (GD 6 to GD 63) at dose levels of 0 (vehicle, 1% aqueous carboxymethylcellulose), 15, 50, or 90 mg/kg body weight at a dose volume of 10 mL/kg body weight. The high dose was chosen as half the lethal dose that was determined in a preliminary study. Clinical signs, body weights, and food consumption were recorded regularly throughout the study. On GD 63, blood samples for hematological examinations, clinical chemistry, and hormone analyses were obtained from all surviving animals by puncturing the retrobulbar venous plexus under isoflurane anesthesia. Thereafter the females were sacrificed and examined macroscopically. Uteri and ovaries were removed. Gravid uterus weights and corrected body weight gain of the dams (terminal body weight minus gravid uterus weight minus body weight on GD 6) were determined. The number of corpora lutea, as well as the number, status, and distribution of implantation sites (live fetuses, early resorptions, late resorptions, dead fetuses) were also determined. Conception rate, preimplantation loss (%), and postimplantation loss (%) were calculated using the litter as unit in the latter two parameters. Placental diameters were measured during gross pathological assessment.

The adrenal glands, liver, and ovaries were weighed. Thereafter, the adrenal glands, liver, ovaries, and placentas including the uterine wall were fixed in 4% buffered formaldehyde solution. After 24–48 hr, the placentas were removed from 4% formaldehyde solution and transferred to 70% ethanol for asservation. All ovaries and placentas were processed for histological examination, stained with hematoxylin-eosin, and examined by light microscopy. For visualization, the placentas were trimmed by a transverse cut through the center (parallel to the umbilical cord) to obtain a section visualizing the different placental zones including the subplacenta. Similarly, the ovaries were trimmed by a longitudinal cut through in the center.

The fetuses were removed from the uterus, sacrificed by subcutaneous injection of pentobarbital (Narcoren®; dose: 0.5 mL/fetus), examined for externally visible anomalies, sexed, and then sectioned. During this final part of the examination, the heart and kidneys were sectioned to evaluate their internal structure. The heads of approximately one-half of the fetuses per litter and of those fetuses that revealed severe externally visible findings were severed from the trunk, fixed in Bouin's solution, processed, and evaluated according to Wilson's method (Wilson, 1965). About 10 transverse sections were prepared per head.

All fetuses were skinned, fixed in ethanol, and stained according to a modified method of Kimmel and Trammell (1981). Thereafter, the stained skeleton of each fetus was examined morphologically referring to the glossary of Wise et al. (1997) and its updated version of Makris et al. (2009). Classifications of findings were based on terms and definitions proposed by the "Berlin Workshops" (Chahoud et al., 1999; Solecki et al., 2001, 2003) and divided into the categories malformations (permanent structural changes, likely adversely affecting survival or health) and variations (changes that may spontaneously occur and considered unlikely to adversely affect survival or health). Fetal findings not classifiable as malformation or variation were described as "unclassified observations."

Pre- and postnatal toxicity study. *Parental (F_0) generation*. Groups of 25 to 27 pregnant animals received epoxiconazole orally by gavage once daily from implantation (GD 6) through 1 day before necropsy (shortly after postnatal day 21) at 0 (vehicle, 1% aqueous carboxymethylcellulose), 15, 50, or 90 mg/kg body weight. The dose volume was 10 mL/kg body weight. Clinical signs, body weights, and food consumption were recorded regularly throughout the study. At necropsy, blood samples for hematological examinations, clinical chemistry, and hormone analyses were obtained from all surviving parental females by puncturing the retrobulbar venous plexus under isoflurane anesthesia.

Thereafter the females were sacrificed and examined macroscopically.

Adrenal glands, brain, kidneys, liver, ovaries, pituitary gland, spleen, thyroid glands (with parathyroid glands), and uteri were weighed and preserved in 4% formaldehyde solution or modified Davidson's solution (ovaries). Additionally, all gross lesions, cervix uteri, oviducts, thyroid glands (including parathyroid glands), and vagina were preserved in 4% formaldehyde solution. The adrenal glands, cervix uteri, ovaries, oviducts, pituitary glands, uterus, vagina, and all gross lesions were processed for histological examination, stained with hematoxylin-eosin, and examined by light microscopy.

F_1 generation. All pups delivered from the F_0 parents (F_1 litter) were examined as soon as possible on the day of birth to determine the total number of pups and the number of live-born and stillborn pups in each litter. All pups, which died before this initial examination, were defined as stillborn pups. From these data, the gestation index, live birth index, postimplantation loss, viability index, and lactation index were calculated. The sex ratio of the pups was determined at birth and confirmed at necropsy.

All pups were observed daily for clinical signs and weighed at birth and weekly thereafter. On postnatal day 21, the pups were sacrificed under isoflurane anesthesia with CO_2. After sacrifice, all pups were examined externally, eviscerated, and their organs assessed macroscopically.

Clinical Pathology

Hematological parameters were measured with an Advia 120 instrument (Bayer, Fernwald, Germany); these comprised red blood cell counts, hemoglobin, hematocrit, mean corpuscular volume, mean corpuscular hemoglobin content, mean corpuscular hemoglobin concentration, reticulocyte, platelet, total white blood cell counts as well as the differential blood cell count.

Clinical chemistry parameters were measured with an Hitachi 917 instrument (Roche, Mannheim, Germany) and consisted of alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, γ-glutamyltransferase, inorganic phosphate, calcium, urea, creatinine, glucose, total bilirubin, total protein, albumin, globulins, triglycerides, and cholesterol.

Steroid hormones were analyzed. Estradiol was measured via ELISA after an ether extraction (extraction efficiency >86%; lowest quantifiable value 2 pmol/L) using a Sunrise MTP-reader, Tecan AG, Maennedorf, Switzerland, and evaluated with the Magellan Software provided by the instrument manufacturer. In addition, the concentrations of 11-deoxycorticosterone, 11-deoxycortisol, 18-hydroxycorticosterone, aldosterone (pre-postnatal study only), androstendione, corticosterone, cortisol, dihydrotestosterone (pre-postnatal study only), progesterone, and testosterone were measured at Metanomics GmbH, Tegeler Weg 33, 10589 Berlin, Germany, by an in-house online solid-phase extraction-LC-MS/MS (SPE-LC-MS/MS) method. Absolute quantification was performed using stable isotope-labeled standards (lowest quantifiable value: 11-deoxycorticosterone, aldosterone, dihydrotestosterone = 0.5 ng/mL; other hormones = 0.1 ng/mL).

Statistics

The data for food consumption, maternal, litter, and post-weaning body weight and body weight change, carcass and gravid uterus weight, duration of gestation, number of corpora lutea and implantations, number of resorptions, number of live fetuses, percent pre- and postimplantation loss, percent live fetuses per litter, number of pups per litter, postimplantation loss, live birth, viability, and lactation indices were analyzed by the two-sided Dunnett's test for the hypothesis of equal means (Dunnett, 1955, 1964). The female mortality, number of pregnant females, and number of litters with fetal findings, were analyzed by pairwise comparison of each dose group with the control group using the one-sided Fisher's exact test for the hypothesis of equal proportions (Siegel, 1956). Gestation index, number of females with live-born and stillborn pups, females with stillborn pups only, stillborn pups, pup mortality data, and number of litters with affected pups at necropsy were analyzed by a one-sided Dunnett's test for the hypothesis of equal means (Siegel, 1956). The proportion of fetuses per litter with findings was analyzed by pairwise comparison of each dose group with the control group by a one-sided Wilcoxon test for the hypothesis of equal medians (Nijenhuis and Wilf, 1978; Miller, 1981; Hettmansberger, 1984). Hematology and clinical chemistry parameters were analyzed by pairwise comparison of each dose group with the control group using a two-sided Kruskall–Wallis test, followed by a two-sided Wilcoxon test for the hypothesis of equal medians in case of $p < 0.05$ (Siegel, 1956). Hormone concentrations were analyzed by pairwise comparison of each dose group with the control group using a two-sided Kruskall–Wallis test, followed by a two-sided Mann–Whitney *U*-test in case of $p < 0.05$ (Siegel, 1956). Organ weights were compared among groups by nonparametric one-way analysis using the two-sided Kruskal–Wallis test, followed by a two-sided Wilcoxon test for the hypothesis of equal medians in case of $p < 0.05$ (Nijenhuis and Wilf, 1978; Miller, 1981; Hettmansperger, 1984).

RESULTS

General

Preparations of the test compound in the vehicle. Homogeneity, stability, and concentration of unlabeled and radiolabeled epoxiconazole were demonstrated for all groups during the toxicokinetic study.

Regarding the prenatal and pre- and postnatal toxicity studies, the stability and homogeneity of the test substance suspensions in the vehicle (1% carboxymethylcellulose suspension in highly deionized water) over a period of 7 days at room temperature was demonstrated. The results of these analyses of the test-substance suspensions also confirmed the correctness of the prepared concentrations. Generally, the analytical values of the samples corresponded to the expected values and were above 90% and below 110% of the nominal concentrations.

Toxicokinetic Study

Plasma radioactivity. The toxicokinetic parameters are summarized in Table 1 and Figure 1.

Table 1
Toxicokinetic Parameters in Pregnant Guinea Pigs after 4 Administrations of Unlabeled Epoxiconazole (GD 6–9) and Radiolabeled Epoxiconazole on GD 10

Dose EPX (mg/kg bw/day)	5	15	50	90
C_{max} (hr)	0.59	1.69	5.04	8.54
T_{max} (hr)	1	1	8	8
$T_{1/2}$ terminal (hr)	45.16	36.75	41.97	35.60
AUC (μg Eq/g×h)	31.91	64.47	352.63	484.45

Both C_{max} and $AUC_{0\text{-}\infty}$ increased with rising dose level in an approximately dose-proportional manner. Terminal elimination half-life values ranged from 36 to 45 hr.

Metabolites. The supernatants from plasma sampled 0.5 to 8 hr after dosing showed metabolite patterns with two main components, the unchanged parent compound epoxiconazole and the metabolite 480M52, accompanied by lower portions of the metabolites 480M13, 480M60, 480M05, 480M54 (or respective isomers), and 480M02 (and its isomer 480M26). In the supernatants from plasma sampled 24 to 96 hr after application, the parent compound was no longer present, and the portions of the metabolite 480M02 and/or its isomer 480M26, and particularly of the metabolite 480M52 had increased.

In all protease solubilizates from the protein precipitates, the metabolite 480M06 represented the main component. 480M06 residue levels increased with the dose level rising from 5 to 50 mg/kg body weight per day, but were relatively lower at 90 mg/kg body weight per day. The parent compound epoxiconazole was only detected in two protease solubilizates in low portions. The metabolite 480M52 was found in both types of plasma fractions (Table 2).

The total concentrations of the parent compound epoxiconazole, the metabolite 480M52, and the sums of the conversion products of epoxiconazole calculated from the two analyzed fractions of all pooled plasma samples are compiled in Table 3.

The concentrations of epoxiconazole in the pooled samples of plasma collected 0.5 to 8 hr after application increased with dose, but the relative portion of epoxiconazole was considerably higher in plasma at 90 mg/kg body weight per day. In plasma sampled 24 to 96 hr after administration, no parent compound was detected any more. These data indicate a limited degradation of epoxiconazole at 90 mg/kg body weight per day during the first 8 hr after administration. The concentrations of the metabolite 480M52 and the sums of all conversion products of epoxiconazole showed an approximately dose-dependent increase.

Metabolic pathway. The proposed metabolic pathway of epoxiconazole following administration by oral gavage to pregnant guinea pigs is shown in Figure 2.

The metabolic pathway of epoxiconazole in pregnant guinea pigs was in good accordance with the pathway reported for pregnant Wistar rats (Thiaener et al., 2011) with the only exception that a peak corresponding to metabolite 480M54 was additionally identified in plasma of pregnant guinea pigs.

Prenatal Toxicity Study

Mortality and clinical signs. Abortion or early delivery occurred in all groups including controls (4, 2, 2, 2 animals in control and ascending dose groups, respectively). Two guinea pigs at 50 and 90 mg/kg body weight per day, respectively, were sacrificed in moribund condition. Two animals at 50 mg/kg body weight per day were found dead, and one animal dosed with 15 mg/kg body weight per day died during administration. In the absence of dose dependency, none of these mortalities were considered to be related to epoxiconazole treatment.

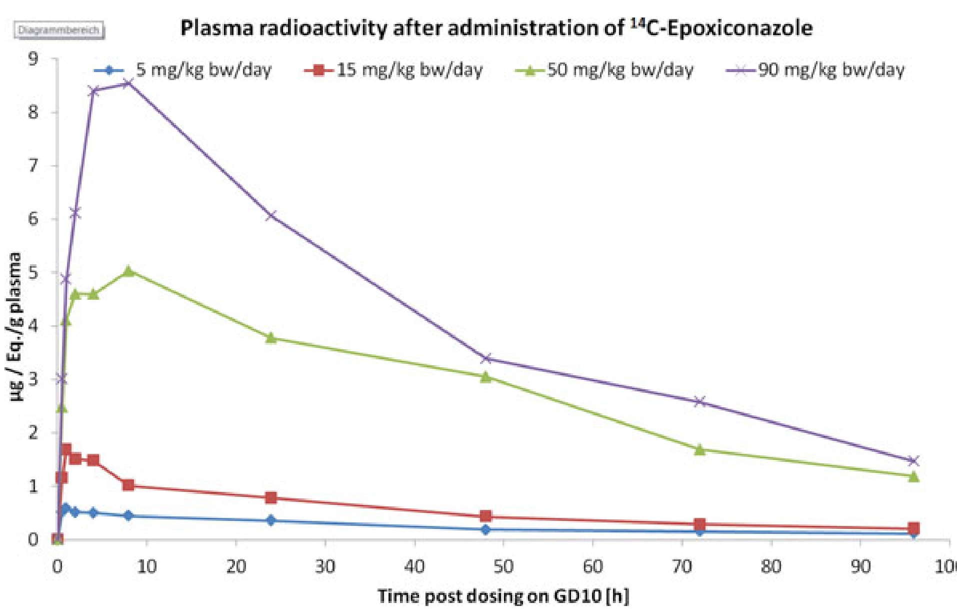

Fig. 1. Plasma radioactivity in pregnant guinea pigs after administration of ^{14}C epoxiconazole.

Table 2
Identification of Radioactive Residues from Pooled Plasma Samples

mg/kg bw/day	Pooled time points	Epoxiconazole µg/mL	Epoxiconazole %TRR	480M52 %TRR	480M13/480M60[a] %TRR	480M05/480M54[b] %TRR	480M02/480M26[c] %TRR
		In supernatants after protein precipitation					
5	0.5–8 hr	0.077	15.4	14.1	4.2	8.2 + 7.5 = 15.7	4.6
15		0.184	11.2	15.7	4.7	8.1 + 10.0 = 18.1	3.3
50		0.501	11.6	13.6	7.0	(6.1 + 1.3) + 9.9 = 17.3	4.7
90		2.604	34.0	15.3	3.3	(5.1 + 1.0 + 0.6) + 7.3 = 14.0	4.1
5	24–96 hr	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
15		n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
50		n.d.	n.d.	30.3	2.1	n.d.	9.3
90		n.d.	n.d.	34.6	n.d.	n.d.	16.1
mg/kg bw/day	Pooled time points	Epoxiconazole µg/mL	Epoxiconazole %TRR	480M52 %TRR	480M06 %TRR	Sum of HPLC peaks eluting 10–28 min %TRR	Sum of HPLC peaks eluting 31–57 min %TRR
		In solubilizates after protease treatment of protein precipitates					
5	0.5–8 hr	n.d.	n.d.	4.7	7.6	12.8	n.d.
15		0.015	0.9	2.4	5.9	7.8	3.4
50		n.d.	n.d.	n.d.	20.5	n.d.	7.8
90		0.130	1.7	2.1	3.8	9.6	0.6
5	24–96 hr	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
15		n.d.	n.d.	10.5	15.5	22.0	n.d.
50		n.d.	n.d.	4.9	29.4	6.5	5.2
90		n.d.	n.d.	4.5	11.5	22.2	6.3

%TRR, radioactivity contained in an individual fraction compared to the total radioactive residue in the particular sample/matrix processed. n.d., not detected; this measurement was below the lower limit of detection.
[a]%TRR for 480M13 or/and isomers or/and 480M60.
[b]%TRR for 480M05/480M54 represents a sum resulting from peaks that were identified in the report either as "480M05 or/and isomers" or "480M05 or/and isomers or/and 480M54."
[c]%TRR for 480M02 or/and its isomer 480M26.

Table 3
Summary of the Total Concentrations of Radioactive Residues, of Unchanged Parent Compound, of Metabolite 480M52 (1,2,4-Triazole) and the Sums of Conversion Products of 14C-Epoxiconazole Detected in Pooled Plasma Samples from Pregnant Guinea Pigs Treated with Epoxiconazole

mg/kg bw/day	Pooled time points	^{14}C residues in plasma (TRR) µg/mL	^{14}C residues in plasma (TRR) %TRR	Epoxiconazole µg/mL	Epoxiconazole %TRR	480M52 µg/mL	480M52 %TRR	Sum of conversion products of epoxiconazole µg/mL	Sum of conversion products of epoxiconazole %TRR
5	0.5–8 hr	0.502	100%	0.077	15.4%	0.094	18.8%	0.394	78.4%
15		1.640	100%	0.198	12.1%	0.296	18.1%	1.149	70.1%
50		4.305	100%	0.501	11.6%	0.587	13.6%	3.876	90.0%
90		7.651	100%	2.734	35.7%	1.333	17.4%	4.971	65.0%
5	24–96 hr	0.198	100%	0.000	0.0%	0.000	0.0%	0.176	89.1%
15		0.534	100%	0.000	0.0%	0.056	10.5%	0.462	86.5%
50		2.522	100%	0.000	0.0%	0.889	35.3%	2.441	96.8%
90		3.658	100%	0.000	0.0%	1.430	39.1%	3.547	97.0%

%TRR, radioactivity contained in an individual fraction compared to the total radioactive residue in the particular sample/matrix processed.

No compound-related clinical signs were observed. Reduced or no defecation occurred in all groups including controls (1, 2, 1, 2 animals in control and ascending dose groups, respectively). One animal at 90 mg/kg body weight per day showed reduced nutritional state, vaginal hemorrhage, and piloerection.

Body weight and food consumption. Body weight development (Suppl. Fig. S1) and food consumption (Fig. S2) were not influenced by the treatment during the study. Likewise, average carcass weight and corrected body weight gain were not influenced by the treatment during the study.

Hematology, clinical chemistry, and hormone measurements. The following significant alterations were observed: decreased red blood cell counts (−6%), hematocrit (−8%), and hemoglobin (−6%) concentrations,

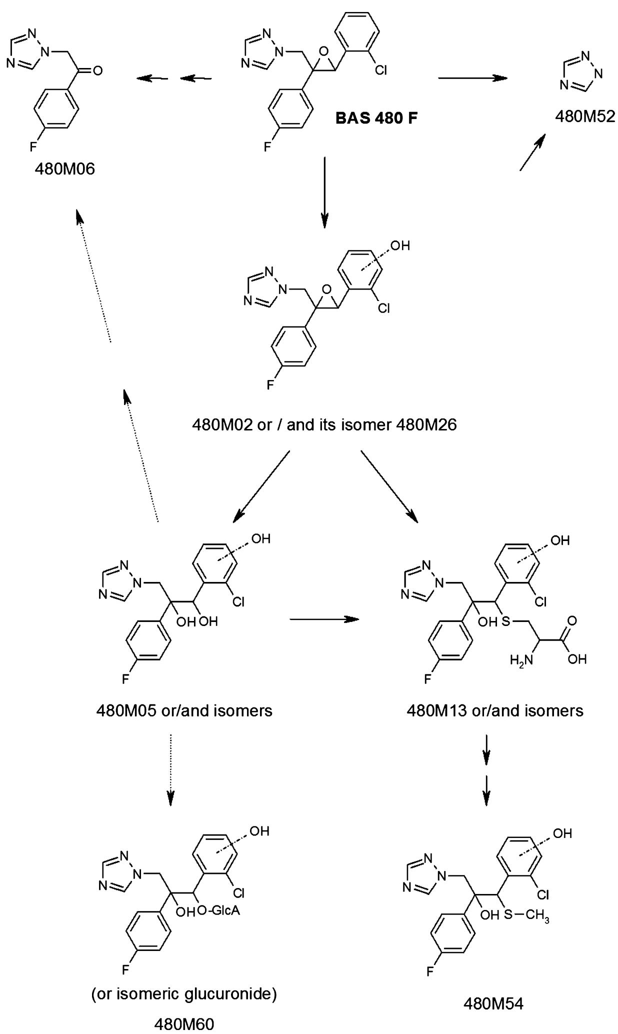

Fig. 2. Metabolic pathway of epoxiconazole in pregnant guinea pigs.

as well as lower total bilirubin and cholesterol levels were observed in guinea pigs at 90 mg/kg body weight per day.

Dose-dependent significant increases in androstenedione, testosterone, and 11-deoxycortisol serum levels were measured in all dose groups. In addition, guinea pigs of the high dose group had significantly higher 18-hydroxycorticosterone and corticosterone serum levels (Table 4). The apparent lower estradiol levels in the treated groups were caused by an artificially high control mean that was generated by just one animal which had a 100-fold higher level than the rest of the group (individual data shown in Table S1). If this outlier was excluded, the means were comparable in all groups: 6.83, 3.08, 4.1, and 5.32 pmol/L.

Cesarean section, necropsy, and histopathology. Gravid uterus weights were not influenced by the treatment (Table 5). No compound-related macroscopic findings were observed at necropsy. The mean number of implantation sites, pre- and postimplantation loss were comparable in all groups. No dead fetuses were noted.

The sex distribution of the fetuses was comparable in all groups. Mean fetal weights did not show any biologically relevant differences between the test substance-treated groups and the control (Table 6).

The absolute and relative adrenal gland weights were slightly, but statistically significantly higher at 90 mg/kg body weight per day. In addition, the mean relative liver weight was significantly higher at both 50 and 90 mg/kg/day, but without dose dependency (Table 7).

A microscopic examination of ovaries and placentas did not reveal any compound-related effect. The diameters of the placentas were comparable across all groups (28.0, 28.6, 27.6, 28.3 mm in control and ascending dose groups, respectively).

Fetal morphology examinations. *Malformations.* Fetal malformations are summarized in Table 8 (for a list of all findings, see Table S2). Umbilical hernia was the only external malformation that was recorded in the fetuses of all dose groups including the controls. Umbilical hernia occurred at litter incidences of 1, 1, 2, and 3 in controls and ascending dose groups, respectively. The presence of this finding in control animals and the low general incidence indicate that there is no evidence of a specific effect of the test substance on complete closure of abdominal wall at the umbilical ring. While the lack of sufficient historical data available in guinea pigs makes assessment difficult, this finding does not appear to be uncommon. A recent reference (Rocca and Wehner, 2009) also describes the presence of omphalocele, a distinct form of umbilical hernia, in four different Dunkin Hartley fetuses from three separate litters of 32 control litters. Another case of omphalocele was reported for one control fetus in a second study from the same group (Wehner et al., 2009). Thus, a relationship of this finding to treatment in the present study seems to be rather unlikely.

No malformations were observed by soft tissue examination.

Malformations of the fetal skeletons were recorded in the fetuses of all groups. For neither the individual findings nor the overall incidences was a dose–response relationship present. Thus, these findings were considered to be spontaneous in nature and without a relation to treatment.

Since the incidence of total external, soft tissue and skeletal malformations showed no dose–response relationship, there was no evidence of a compound-related effect.

Variations. Fetal variations are summarized in Table 9 (for a list of all findings, see Table S3). No variations were observed by external examination of the fetuses.

Variations observed by soft tissue examinations consisted of malpositioned carotid origin and were detected in the low- and mid-dose groups as well as the control group. As there was no dose–response relationship, these findings were considered to be spontaneous in nature and without a relation to treatment.

Variations in different skeletal structures were detected with or without effects on the corresponding cartilage. Skeletal variations consisted primarily of minor delays or disturbances of ossification. The majority of skeletal variations were equally distributed among the different groups. However, two skeletal variations, a fusion of the thoracic centrum and thoracic arch and a misshapen

Table 4
Prenatal Toxicity Study: Hormone Concentrations in Guinea Pigs, Gestation Day 63 (Individual Values, See Table S1)

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
Number of samples	23	20	18	18
11-Deoxycorticosterone (nmol/L)	250 ± 104	220 ± 123	220 ± 100	255 ± 100
18-Hydroxycorticosterone (nmol/L)	0.47 ± 0.23	0.48 ± 0.17	0.55 ± 0.28	0.73 ± 0.31**
Androstenedione (nmol/L)	9.09 ± 3.76	14.26 ± 7.35*	17.05 ± 9.48**	16.93 ± 7.63**
Corticosterone (nmol/L)	62.38 ± 36.63	62.89 ± 23.60	76.57 ± 41.00	89.70 ± 27.80**
Cortisol (nmol/L)	8300 ± 2419	7954 ± 2020	8214 ± 2647	8682 ± 2100
Estradiol (pmol/L)	18.70 ± 57.98	3.08 ± 0.13	4.10 ± 0.45	5.32 ± 14.63
Progesterone (nmol/L)	820 ± 299	889 ± 287	876 ± 286	799 ± 242
Testosterone (nmol/L)	0.63 ± 0.19	0.89 ± 0.40*	1.12 ± 0.34**	1.25 ± 0.41**
11-Deoxycortisol (nmol/L)	34.90 ± 14.35	50.39 ± 23.64*	75.71 ± 63.03**	126.26 ± 99.90**

*$p \leq 0.05$; **$p \leq 0.01$ [Kruskal–Wallis test plus Wilcoxon test (estradiol) or Mann–Whitney U-test (other hormones)]. Means ± SD are shown.

Table 5
Prenatal Toxicity Study: Gravid Uterus Weights and Maternal Body Weight Change

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
N	23	22	19	19
Gravid uterus (g)	417.8 ± 146.70	401.2 ± 142.39	393.1 ± 91.23	392.1 ± 158.31
Carcass (g)	858.4 ± 123.20	865.5 ± 62.33	883.9 ± 113.73	852.9 ± 84.43
Net weight change from day 6 (g)	41.0 ± 107.56	52.2 ± 64.80	58.3 ± 69.27	30.3 ± 68.16

*$p \leq 0.05$; **$p \leq 0.01$ (two-sided Dunnett test). Means ± SD are shown.

sternebra were noted at statistically significantly higher incidences.

Early ossification of the thoracic vertebral centrum and neural arches is likely to represent an advancement in the ossification and development of the vertebrae. These changes should occur normally in the ossification of the vertebrae as development proceeds postnatally. A cartilaginous joint between the centrum and vertebral arches is present at birth in humans, but the cartilaginous regions ossify allowing the vertebra (consisting of a body and a neural arch) to become a single bone around 3 to 6 years of age (Moore and Persaud, 2008). This process is similar in rats and rabbits and should be the same in the guinea pig, but we could not find specific information regarding the timing of the process in this species. Because this fusion is part of the normal process of vertebral development, it is unclear why these changes would have occurred more frequently as the dose increased in this study. The weight of fetuses in each group was similar and the litter sizes, though variable, were similar across all treatment groups, both of which can influence the degree of ossification. It is probable that this process begins around the time of birth in guinea pigs (fusions were noted in seven control fetuses); if so, there is likely to be a good deal of variability in the timing of ossification of the connection between the centrum and vertebral arches. This type of variability can be seen in several bones of rat fetuses that are ossifying around the time of birth, for example, the sternebrae, vertebral centra, and the carpals/tarsals, metacarpals/metatarsals, and phalanges. One may conclude that these minor changes in the ossification process, which occur very frequently in rat and rabbit fetuses, seem also to be common skeletal abnormalities in guinea pigs. Rocca and Wehner (2009) reported skeletal abnormalities such as short rib, discontinuous rib, supernumerary rib, unossified vertebra, bipartite ossification, and extra sites of ossification in 35 to 50% of litters (9.6–24.7% of fetuses) examined. An increase of this spontaneously high incidence of skeletal variations is often noted in the presence of maternal toxicity or maternal stress, as has been substantiated in this study for the high-dose females displaying anemia and increased adrenal weights, as well as increased 18-hydroxycorticosterone and corticosterone serum levels.

Taking all this into consideration, the increased skeletal variations in this study and the subsequently increased total incidence of variations were regarded to be of no toxicological relevance and were consequently not classified as adverse events.

Unclassified findings. Unclassified findings are summarized in Table 10 (for a list of all findings, see Table S4). Unclassified findings consisted of blood coagulum around placenta (control), discolored kidneys (three fetuses of the low-dose group), and isolated cartilage findings without influence on the respective bone structures in all groups including the control. Unclassified cartilage findings were exclusively related to the sternum and the ribs. They were significantly increased in all treated groups but did not exhibit a specific pattern or a dose response.

Pre- and Postnatal Toxicity Study

In Life Observations.

Parental generation. There were no test substance-related mortalities in any of the female F_0 parental animals in the low- and mid-dose groups. One high-dose female was found dead on gestation day 52 after showing piloerection and vaginal discharge. Necropsy revealed

Table 6
Prenatal Toxicity Study: Summary of Reproduction Data

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Females mated	*N*	30	30	30	30
Pregnant	*N*	27	26	25	23
Conception rate	%	90	87	83	77
Aborted	*N*	3	1	1	0
Premature births	*N*	1	1	1	2
Dams with viable fetuses	*N*	23	21	19	18
Dams with total litter loss	*N*	0	1	0	1
Female mortality[F]	*N*	3	3	5	2
	%	10	10	17	7
Pregnant at terminal sacrifice	*N*	23	22	19	19
	%	77	73	63	63
Corpora lutea[D]	Mean ± SD	5.7 ± 1.47	5.6 ± 1.53	5.7 ± 1.33	5.8 ± 1.13
Implantation sites[D]	Mean ± SD	4.7 ± 1.72	4.4 ± 1.53	4.7 ± 1.06	4.8 ± 1.40
Preimplantation loss (%)[D]	Mean ± SD	18.4 ± 20.33	21.4 ± 21.70	16.4 ± 17.94	16.9 ± 20.21
Postimplantation loss (%)[D]	Mean ± SD	12.1 ± 17.52	8.8 ± 13.47	10.7 ± 17.83	14.2 ± 26.25
Early resorptions (%)[D]		9.7 ± 16.69	8.8 ± 23.47	7.2 ± 17.08	13.2 ± 26.41
Late resorptions (%)[D]		2.4 ± 6.31	0	3.5 ± 6.98	1.1 ± 4.59
Dead fetuses		0	0	0	0
Dams with viable fetuses	*N*	23	21	19	18
Live fetuses[D]	Mean ± SD	4.2 ± 1.83	4.1 ± 1.28	4.2 ± 1.01	4.3 ± 1.56
Live fetuses (%)[D]	Mean ± SD	87.9 ± 17.52	95.5 ± 11.96	89.3 ± 17.83	90.5 ± 16.55
Males[D]	Mean ± SD	2.2 ± 1.44	2.1 ± 1.15	1.9 ± 1.18	2.1 ± 1.45
Males (%)[D]	Mean ± SD	45.0 ± 27.22	50.9 ± 25.88	42.6 ± 24.64	45.4 ± 25.01
Females[D]	Mean ± SD	2.0 ± 1.13	2.0 ± 1.10	2.2 ± 1.27	2.2 ± 1.34
Females (%)[D]	Mean ± SD	42.9 ± 24.66	44.6 ± 25.63	46.7 ± 26.44	45.1 ± 27.70
% Males		52.1	51.7	46.8	49.4
Fetal weights[D]	Mean ± SD	80 ± 13.1	81 ± 10.9	74 ± 10.1	76 ± 7.6
	N	23	21	19	18
Males[D]	Mean ± SD	79 ± 16.3	78 ± 12.3	74 ± 12.6	79 ± 8.0
	N	20	20	17	18
Females[D]	Mean ± SD	80 ± 13.0	82 ± 12.7	72 ± 11.5	71 ± 9.0
	N	20	19	18	16

*$p \leq 0.05$; **$p \leq 0.01$ (F, one-sided Fisher test; D, two-sided Dunnett test).

Table 7
Prenatal Toxicity Study: Mean Absolute and Relative Organ Weights

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
N	26	25	24	26
Terminal body weight (g)	864.7 ± 118.9	858.6 ± 62.0	876.3 ± 104.3	835.2 ± 88.4
Adrenal weight (mg)	632.8 ± 132.3	606.1 ± 104.1	665.9 ± 151.7	698.0 ± 101.5*
Liver weight (g)	23.9 ± 3.5	24.8 ± 3.1	26.8 ± 4.8	25.0 ± 3.8
Ovary weight (mg)	266.6 ± 140.6	369.8 ± 156.4	263.0 ± 177.7	227.5 ± 141.0
Adrenal weight (%)	0.075 ± 0.016	0.071 ± 0.012	0.076 ± 0.014	0.084 ± 0.014*
Liver weight (%)	2.827 ± 0.47	2.894 ± 0.3	3.048 ± 0.281**	2.991 ± 0.279*
Ovary weight (%)	0.031 ± 0.017	0.031 ± 0.017	0.029 ± 0.017	0.027 ± 0.015

*$p \leq 0.05$; **$p \leq 0.01$ (two-sided Kruskal–Wallis and Wilcoxon test). Means ± SD are shown.

that it died most likely while it aborted. Another high-dose female was sacrificed moribund on gestation day 71 (during parturition) after showing hypothermia, piloerection, and lateral position. No other compound-related clinical signs were observed throughout the gestation and lactation period. One control animal, one low-dose female, and one high-dose female were sacrificed prematurely after abortion on gestation days 46, 57, and 27, respectively.

At 90 mg/kg body weight per day, the body weights were approximately 6% below control values at the end of gestation, although the difference was not statistically significant. Body weight gain of the high-dose females was decreased by approximately 15% during gestation. Body weight gain was not affected by epoxiconazole at 15 and 50 mg/kg body weight per day. Body weight gain was not influenced by the treatment during lactation in all groups (Fig. S3).

Food consumption of the high-dose females was statistically significantly below control values (12–13%) during the last week of gestation. Food consumption of the low- and mid-dose females was comparable to the control animals during premating, gestation, and lactation (Fig. S4).

Table 8
Fetal Morphology Data: Salient Malformations (for Complete Dataset See Table S2)

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Litters evaluated	*N*	23	21	19	18
Fetuses evaluated	*N*	96	87	79	77
Umbilical hernia					
Fetal incidence	*N*	1	1	2	4
	%	1.0	1.1	2.5	5.2
Litter incidence[F]	*N*	1	1	2	3
	%	4.3	4.8	11	17
Affected fetuses per litter[W]	Mean% ± SD	0.9 ± 4.17	0.8 ± 3.64	2.4 ± 7.14	4.3 ± 10.71
Total fetal malformations					
Fetal incidence	*N*	7	8	5	9
	%	7.3	9.2	6.3	12
Litter incidence[F]	*N*	6	7	4	6
	%	26	33	21	33
Affected fetuses per litter[W]	Mean% ± SD	7.1 ± 13.51	7.7 ± 11.95	6.5 ± 13.25	13.4 ± 23.95

*$p \le 0.05$; **$p \le 0.01$ (F, one-sided Fisher's exact test; W, one-sided Wilcoxon).

Table 9
Fetal Morphology Data: Salient Variations (for Complete Dataset See Table S3)

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Litters evaluated	*N*	23	21	19	18
Fetuses evaluated	*N*	96	87	79	77
Malpositioned carotid origin					
Fetal incidence	*N*	5	2	1	0
	%	5.2	2.3	1.3	0.0
Litter incidence[F]	*N*	5	2	1	0
	%	4.3	0	0	0
Affected fetuses per litter[W]	Mean% ± SD	5.6 ± 12.44	1.9 ± 6.02	1.1 ± 4.59	0
Thoracic centrum fused with arch					
Fetal incidence	*N*	7	19	23	25
	%	7.3	22	29	32
Litter incidence[F]	*N*	6	11	12*	13**
	%	26	52	63	72
Affected fetuses per litter[W]	Mean% ± SD	6.6 ± 13.04	21.7 ± 28.77*	26.3 ± 29.24**	38.8 ± 34.58**
Misshapen sternebra; unchanged cartilage					
Fetal incidence	*N*	5	11	6	12
	%	5.2	13	7.6	16
Litter incidence[F]	*N*	4	7	5	9*
	%	17	33	26	50
Affected fetuses per litter[W]	Mean% ± SD	6.6 ± 17.27	11.8 ± 20.51	7.1 ± 12.73	18.8 ± 26.20*
Total fetal variations					
Fetal incidence	*N*	45	56	55	56
	%	47	64	70	73
Litter incidence[F]	*N*	21	20	17	18
	%	91	95	89	100
Affected fetuses per litter[W]	Mean% ± SD	51.8 ± 31.16	68.1 ± 30.24*	67.2 ± 32.11*	75.5 ± 21.26**

*$p \le 0.05$; **$p \le 0.01$ (F, one-sided Fisher's exact test; W, one-sided Wilcoxon).

Table 10
Fetal Morphology Data: Unclassified Observations (for Complete Dataset See Table S4)

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Litters evaluated	*N*	23	21	19	18
Fetuses evaluated	*N*	96	87	79	77
Total fetal unclassified cartilaginous observations					
Fetal incidence	*N*	2	13	6	8
	%	2.1	15	7.6	10
Litter incidence[F]	*N*	1	6*	5	5*
	%	4.3	29	26	28
Affected fetuses per litter[W]	Mean% ± SD	1.4 ± 6.95	12.9 ± 25.58*	7.6 ± 14.28*	17.0 ± 34.32*

*$p \le 0.05$; **$p \le 0.01$ (F, one-sided Fisher's exact test; W, one-sided Wilcoxon).

Table 11
Pre- and Postnatal Toxicity Study: Gestation and Delivery Parameters

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Females mated	*N*	27	25	27	27
Pregnant	*N*	27	25	27	27
Without delivery	*N*	1	1	0	3
Delivering	*N*	26	24	27	24
	%	96.3	96.0	100.0	88.9
With live-born pups	*N*	26	24	27	24
	%	100.0	100.0	100.0	100.0
With stillborn pups	*N*	1	2	0	3
	%	3.8	8.3	0	12.5
With stillborn pups only	*N*	0	0	0	0
	%	0	0	0	0
Gestation index (%)	%	100.0	100.0	100.0	100.0
Gestation length (days)	Mean ± SD	69 ± 1	69 ± 1	69 ± 1	69 ± 1
	N	26	24	27	24

$^{*}p \leq 0.05$; $^{**}p \leq 0.01$ (one-sided Dunnett test).

Mean duration of gestation was 69 days in all test groups. All gestational parameters and indices were comparable between control and test substance treated groups (Table 11). Thus, epoxiconazole did not adversely affect gestation and delivery by the F_0 generation parental females.

Offspring. The number of delivered pups and average litter size was statistically comparable in all groups (Table 12). However, the number of live-born pups and subsequent numbers during lactation were slightly lower at 90 mg/kg body weight per day, attributable to two decedents and one abortion in this dose group. This effect was considered to be a result of systemic maternal toxicity. The correlation to maternal toxicity was evident since neither the average number of implants nor postimplantation loss, gestation, and live birth indices were significantly influenced by the high dose of the test material. Rates of live-born and stillborn pups were evenly distributed among the groups.

The test substance did not influence pre-weaning pup survival in any of the treated groups. There were no test substance-related clinical findings in any of the dose groups. Likewise, the body weight development of pups remained unaffected by administration of epoxiconazole (see Table S5).

Postmortem examinations. There were no compound-related effects on hematological and clinical chemistry parameters (data not shown). Progesterone and 21-hydroxyprogesterone levels were significantly higher at 90 mg/kg body weight per day. 21-Hydroxyprogesterone concentrations were also significantly higher at 50 mg/kg body weight per day (Table 13). Androstenedione, aldosterone, dihydrotestosterone, and testosterone concentrations were below the limit of quantification in nearly all samples.

The uterus weights were lower in all dose groups. However, in absence of a dose dependency and histopathological changes these differences were considered not to be compound related. Relative liver weights were increased at 90 mg/kg body weight per day, and relative adrenal weights were increased at 50 and 90 mg/kg body weight per day. All other weight parameters were comparable to controls (Table 14).

Macroscopically, yellow, white, or red foci were observed in the liver of 1 of 25 animals at 15 mg/kg body weight per day, 6 of 27 animals at 50 mg/kg body weight per day, and 5 of 27 animals at 90 mg/kg body weight per day, respectively. A few ovarian cysts ranging from 1 to 7 mm in diameter were evident in the control group compared to those observed in the 90 mg/kg body weight per day group, in which cysts were macroscopically visualized in only 1 of 27 animals.

Microscopic examinations of the adrenal glands revealed multifocal areas of cytoplasmic change in the deep zona fasciculata increasing in incidence and severity from 50 mg/kg body weight per day to 90 mg/kg body weight per day (Table 15). In these areas, the cells of the zona fasciculata lost their large vacuoles and displayed a slight basophilic and foamy appearance. Other cells contained small, needle-shape cytoplasmic crystals, resembling cholesterol crystals. In addition, activated sinusoidal cells and minimal mononuclear infiltrating cells were also observed. Vacuolation of the zona fasciculata was increased in the 50 and 90 mg/kg body weight per day groups.

In the liver (only livers with macroscopic findings were examined), macroscopic foci correlated with acute focal coagulation necrosis accompanied by minimal or no inflammatory response and occasional, minimal dystrophic mineralization was observed. Since no signs of hepatotoxicity were found in the clinical chemistry of these animals, the acute necrosis was attributed to the isoflurane anesthesia performed for the blood sampling before the animals were sacrificed for necropsy. (Hursh et al., 1987; Clarke et al., 1995; Durak et al., 1999; Percy and Barthold, 2001).

Ovarian cysts visualized macroscopically in control animals correlated with cystic rete ovarii, a common finding in guinea pigs (Keller et al., 1987; Shi et al., 2002). Unlike follicular or luteal cysts, rete ovarii cystic structures lack follicular or luteal cells and display single ciliated cells in their epithelium that ranges from columnar to flattened depending on the dilation grade.

Table 12
Pre- and Postnatal Toxicity Study: Litter Parameters

Dose EPX (mg/kg bw/day)		0 (Vehicle)	15	50	90
Pups delivered	*N*	85	75	80	65
	Mean ± SD	3.3 ± 1	3.1 ± 1	3.0 ± 1	2.7 ± 1
Pups live-born	*N*	84	73	80	62
	%	98.8	97.3	100.0	95.4
Pups stillborn	*N*	1	2	0	3
	%	1.2	2.7	0.0	4.6
Pup weight at birth (g)	Mean ± SD	100.9 ± 9.6	101.7 ± 13.1	101.2 ± 10.4	96.3 ± 11.2
Viability index	%	98.1	100.0	100.0	95.8
Weaning index	%	100.0	100.0	98.5	99.2

*$p \leq 0.05$; **$p \leq 0.01$ (two-sided Dunnett test).

Table 13
Pre- and Postnatal Toxicity Study: Serum Hormone Concentrations at End of Weaning

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
Number of samples	25	24	27	24
Progesterone (nmol/L)	4.56 ± 5.15	6.65 ± 6.24	7.17 ± 6.16	8.41 ± 5.39**
21-Hydroxyprogesterone (nmol/L)	0.67 ± 0.58	0.71 ± 0.41	0.97 ± 0.62*	1.23 ± 0.61***
18-Hydroxycorticostereone (nmol/L)	1.67 ± 0.94	1.35 ± 0.46	1.50 ± 0.75	1.29 ± 0.54
11-Deoxycortisol (nmol/L)	1.63 ± 1.42	1.35 ± 1.58	1.25 ± 0.73	1.79 ± 1.68
Corticosterone (nmol/L)	8.54 ± 8.81	6.23 ± 8.71	6.06 ± 4.14	7.77 ± 5.62
Cortisol (nmol/L)	1533 ± 826	1382 ± 590	1325 ± 513	1366 ± 792
Estradiol (pmol/L)	1.13 ± 2.45	1.12 ± 2.71	0.88 ± 1.83	4.15 ± 15.85

*$p \leq 0.05$; **$p \leq 0.01$ [Kruskal–Wallis test plus Wilcoxon test (estradiol) or Mann–Whitney *U*-test (other hormones)]. Means ± SD are shown.

Table 14
Pre- and Postnatal Toxicity Study: Organ Weights

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
N	26	24	27	24
Terminal body weight (g)	735.5 ± 57.2	727.8 ± 59.8	715.4 ± 59.8	736.3 ± 72.2
Adrenal weight (mg)	476.0 ± 91.0	504.6 ± 92.1	520.0 ± 78.3	509.4 ± 57.5
Liver weight (g)	23.0 ± 3.9	22.4 ± 4.5[a]	23.3 ± 3.6	25.2 ± 3.4
Uterus weight (g)	1.959 ± 0.602	1.651 ± 0.587*	1.727 ± 0.505	1.617 ± 0.659**
Adrenal weight (%)	0.065 ± 0.012	0.070 ± 0.011	0.073 ± 0.011**	0.070 ± 0.009*
Liver weight (%)	3.120 ± 0.45	3.109 ± 0.416	3.251 ± 0.379	3.423 ± 0.385**
Uterus weight (%)	0.269 ± 0.089	0.229 ± 0.074[a]	0.242 ± 0.068	0.218 ± 0.077**

*$p \leq 0.05$; **$p \leq 0.01$ (two-sided Kruskal–Wallis H and Wilcoxon test). Means ± SD are shown.
[a]$N = 23$.

One control animal, one low-dose female, and one high-dose female were sacrificed prematurely after abortion (on GD 46, 57, and 27, respectively). One high-dose F_0 female died intercurrently on gestation day 52, most likely while it aborted, as was obvious by the retention of a dead fetus in its vagina. Another high-dose animal was sacrificed on GD 71 in a moribund state, showing hypothermia, piloerection, and lateral position, because it seemed to be unable to deliver. All of these animals showed similar alterations in the uterus, cervix, and vagina. Namely, the high-dose (90 mg/kg body weight per day) animal that died on GD 52 showed necrosis, hemorrhage, and severe inflammatory changes mainly affecting the uterus. The other animal of this dose group that was sacrificed on GD 71 showed similar, but less severe, changes predominantly in the cervix. In both animals, large congestive mesometrial vessels invaded by trophoblast cells were observed in the uterus and in the oviducts. All other animals with abortion showed similar characteristics, the uterus being more severely affected (hemorrhage, necrosis, and presence of trophoblast cells) than the vagina (edema, congestion, or infiltrates).

DISCUSSION

Epoxiconazole was well absorbed after oral administration, as shown by dose dependently increased C_{max} and $AUC_{0-\infty}$ values. Observed plasma ^{14}C equivalents indicated comparable exposures of guinea pigs when compared to rats treated with comparable dose levels of epoxiconazole (Fabian and Landsiedel, 2011). At the doses of 5 and 50 mg/kg body weight per day, $AUC_{0-\infty}$ values were 30.35 and 284.11 μg Eq/g·h in rats and 31.91 and 352.63 μg Eq/g·h in guinea pigs, respectively. Therefore, the species

Table 15
Pre- and Postnatal Toxicity Study: Incidences of Microscopic Changes in Adrenals

Dose EPX (mg/kg bw/day)	0 (Vehicle)	15	50	90
Number of animals examined	27	25	27	27
Cytoplasmic change, (multi)focal	0	0	10	14
Grade 1	0	0	8	3
Grade 2	0	0	2	11
Vacuolation, cytoplasmatic, diffuse	27	25	27	27
Grade 1	1	1	0	3
Grade 2	6	6	2	0
Grade 3	16	14	7	11
Grade 4	4	4	18	13

differences observed in the present studies cannot be explained by differing epoxiconazole absorption between guinea pigs and rats.

Epoxiconazole was administered orally to pregnant guinea pigs daily either from implantation to shortly before the expected day of parturition, or from implantation throughout gestation and 3 weeks weaning, at doses up to 90 mg/kg body weight per day. The only effect at 90 mg/kg body weight per day was a slightly higher number of decedents due to the abortion seen in the pre- and postnatal study. No effect on body weight and food consumption was seen in the prenatal toxicity study, whereas a slight decrease of these parameters was evident in the pre- and postnatal toxicity study.

Regarding clinical pathology, a mild anemia, indicated by decreased erythrocyte counts, hemoglobin, and hematocrit values, was present at 90 mg/kg body weight per day in the prenatal toxicity study. More pronounced anemia was observed in previous experiments with rats. However, no effects on hematological and clinical chemistry parameters were observed at the end of weaning in the pre- and postnatal toxicity study.

At the end of gestation, the higher androgen levels in guinea pigs of all dose groups are probably a consequence of aromatase inhibition. At this time, decreased estradiol levels could hardly be detected because many of the individual animal's estradiol levels were below the detection limit of the assay. The higher 18-hydroxycorticosterone and corticosterone levels at 90 mg/kg body weight per day and increased 11-deoxycortisol levels in all dose groups may be a result of alterations in steroid hormone production of the adrenals. At the end of weaning, higher progesterone and 21-hydroxyprogesterone levels were present at 90 mg/kg body weight per day. 21-Hydroxyprogesterone concentrations were also higher at 50 mg/kg body weight per day than in controls.

Liver weights were slightly increased at 90 mg/kg body weight per day, possibly due to the adaptation to increased metabolic load. Slight increases in adrenal gland weights were observed in both studies at 50 and 90 mg/kg body weight per day. This weight increase correlated with increased 18-hydroxycorticosterone and corticosterone levels at the end of gestation and to vacuolization of the zona fasciculata observed at end of weaning. These effects might reflect either an alteration in the steroid hormone production of the adrenals or potential consequences of maternal stress at 50 and 90 mg/kg body weight per day. It is noteworthy, that in contrast to what was observed in rats, the histopathology of all guinea pig placentas revealed no adverse effect caused by epoxiconazole up to and including a dose of 90 mg/kg body weight per day.

Evaluation of the cesarean section parameters revealed no test substance-related, adverse effects on the mean number of corpora lutea, implantation sites, pre- and postimplantation loss, numbers of live fetuses, fetal weights, and sex distribution. No dead fetuses were noted. Likewise, pre- and postnatal administration of epoxiconazole did not affect gestation or lactation of the parental females up to and including a dose of 90 mg/kg body weight per day. In particular, gestation length, parturition, lactation, and weaning as well as sexual organ weights and gross and histopathological findings of these organs were comparable between guinea pigs of all test groups. There was no effect on pup viability or pup body weights.

Examination of the fetuses revealed a slightly higher rate of mainly skeletal variations in all dose groups. Such slight retardations of the ossification process occur very frequently in rat and rabbit fetuses and they seem also to be common skeletal abnormalities in guinea pigs as previously experienced by Rocca and Wehner (2009). Taking this and the potential consequences of maternal stress at the high dose into consideration, the increased skeletal variations in this study and the subsequently increased total incidence of variations were regarded to be of no toxicological relevance and are not classified as adverse events, although the limited historical data available are a significant obstacle to a thorough interpretation of these findings in guinea pigs.

It is important to emphasize that test substance-related, specific adverse effects on fetal morphology were not observed in this study, at the tested dose levels up to 90 mg/kg body weight per day. Furthermore, the observed minor abnormalities do not form a pattern, which would be indicative of the well-known triazole-related craniofacial malformations described for murid rodents, such as cleft palate, hypognathia, hydrocephaly, exophthalmus, macroglossia, and clusters of variations in the skull bones (Menegola et al., 2006).

Relevance of Findings for Humans

Epoxiconazole was tested in guinea pigs for prenatal and pre-postnatal toxicity up to a dose level of 90 mg/kg body weight per day. The high-dose level was based on the results of a range-finding study in pregnant guinea pigs, in which the dose of 180 mg/kg body weight caused mortality in 7 of 20 animals and caused distinct signs of clinical toxicity in the remaining animals, such as abortion, poor general state, as well as markedly reduced food consumption and loss of body weight (Schneider et al., 2011). In the present studies there were no compound-related malformations, no increased incidence of intrauterine deaths, and no adverse effects on parturition, postnatal growth and development. Furthermore, no compound-related effects were observed by histological examination of placentas. Thus, the results in guinea pigs are in profound contrast to those obtained in epoxiconazole-treated rats, which showed dystocia, as well as increased incidence of late postimplantation loss associated with degeneration of placental labyrinth and trophospongium at the dose level of 50 mg/kg body weight per day. The fact that maternal serum estradiol was markedly reduced, and that the adverse effects could be prevented by co-administration of estradiol, clearly demonstrates that sufficient levels of this hormone are crucial for proper fetal development in this species. Due to its general toxicity in guinea pigs, epoxiconazole could not be tested at 180 mg/kg body weight per day, a dose level that induced fetal malformations in rats.

Markedly decreased estradiol levels were observed in rats after administration of epoxiconazole. Estradiol levels were not influenced in guinea pigs. However, observed increases of the estradiol precursors testosterone and androstenedione suggest that guinea pigs are principally sensitive to aromatase inhibition by epoxiconazole.

Based on the results of the present studies it is evident that guinea pigs react quite differently to epoxiconazole treatment when compared to rats. Therefore, it is important to assess the human relevance of the results obtained in rats or guinea pigs. Based on the nature of the effects observed, the focus is set on the comparison of pregnancy and parturition in murid rodents, guinea pigs, and primates including humans.

Regarding pregnancy and parturition, guinea pigs have a number of similarities to humans regarding hormonal regulation (Batra et al., 1980; Hobkirk and Glasier, 1993; Furukawa et al., 2011), placentation (Carter, 2007), trimester-equivalent gestation (Trewin and Hutz, 1998), extensive prenatal organ system development (Nishimura and Shiota, 1977), including the brain growth spurt (Dobbing and Sands, 1979) and parturition (Mitchell and Taggart, 2009).

Placenta. The guinea pig placenta share similarities with both humans and murid rodents. As in humans, the guinea pig placenta is hemomonochorial (Enders, 1965; Carter and Enders, 2004; Carter, 2007) and the trophoblast invades extensively into the endometrium. Murid placentas are hemotrichorial, and invasion of the trophoblast is rather shallow. On the other hand, like murid rodents, guinea pigs retain the yolk sac placenta until term. Guinea pigs possess a subplacenta not present in humans or rats, which does not contribute to maternal-fetal transfer.

Pregnancy. Guinea pigs deliver well-developed (precocial) offspring after a relatively long gestation period. Thus many events occurring during human fetal development also occur during fetal life in guinea pigs. In contrast, rodents have a short gestation period and deliver rather immature (altricial) pups, where many of these processes occur during postnatal development. Therefore, guinea pigs are more similar to humans with respect to relative pregnancy duration and intrauterine development. As a species for prenatal toxicity testing, guinea pigs are considered to be a predictable species regarding detection of teratogenic effects (Schardein et al., 1985). Among other agents, guinea pigs are sensitive to retinoid acid-induced malformations and are therefore potentially susceptible to the proposed mechanism of triazole-mediated teratogenicity (Menegola et al., 2006).

Hormonal regulation of pregnancy and parturition. In rats and mice, corpora lutea, which are active throughout gestation, are the source of progesterone. Therefore, the presence and activity of corpora lutea is absolutely necessary for the maintenance of gestation. Mating induces a prolactin surge in the pituitary, which is necessary for sustained corpora lutea activity. Subsequently, luteotrophic lactogens synthetized in the trophoblast bind to prolactin receptors of corpora lutea and maintain progesterone secretion throughout gestation (Malassiné et al., 2003). In contrast, the maintenance of gestation in guinea pigs is only dependent on the activity of corpora lutea during the first 4 weeks. Thereafter, progesterone is synthesized by the placenta (Csapo et al., 1981). This is analogous to humans, in which the corpus luteum is active only during the first trimester. Thereafter placental progesterone production maintains pregnancy, even after ovarectomy. Thus, endocrine mechanisms maintaining pregnancy are quite similar in humans and guinea pigs, but fundamentally different from murid rodents.

Ovaries are the source of estrogen in nonpregnant and pregnant rats. In contrast, pregnant guinea pigs and pregnant humans produce estrogens throughout the major part of pregnancy mainly via the placenta (Batra et al., 1980; Hobkirk and Glasier, 1993). This estrogen synthesis in human placenta is dependent on the synthesis of androgen precursors (dehydroepiandrostenedione) in maternal and fetal adrenals (feto-placental unit). The complete synthesis of estrogen in human placenta is not possible due to absence of steroid 17α-hydroxylase (Strauss et al., 1996; Fuchs and Fields, 1998; Furukawa et al., 2011). In contrast, the rat placenta possesses 17α-hydroxylase, but lacks aromatase.

In humans and guinea pigs neonates are born at a much more advanced stage of maturity, as compared to rodents. Parturition in humans and guinea pigs is considered to be a multifactorial process, which is not as precisely regulated as in rodents. This is evidenced by the occurrence of "spontaneous" preterm births, which are common in humans and guinea pigs, but rare in rats and mice (Mitchell and Taggart, 2009).

In rats, expression of oxytocin receptors is dependent on an increase in estrogen and a strong decrease in progesterone secretion (progesterone withdrawal) during preparation for parturition. In contrast, a drop in progesterone levels does not occur in guinea pigs and primates. Rising levels of estrogen are sufficient for the induction of oxytocin receptors in these species (Fuchs and Fields, 1998). Preterm luteolysis is also not required in guinea pigs, higher primates, and humans; this is in striking contrast to what is observed in murid rodents (Challis et al., 1975; Thorburn and Challis, 1979). Steroid hormone levels during pregnancy, as well as responses to antigestins, are also similar in guinea pigs and humans (Challis et al., 1971; Tulchinsky et al., 1972; Elger and Hasan, 1985). In guinea pigs, prostaglandins are responsible for the initiation of labor. PGE_2 and PGF2α are synthesized in amnion, yolk sac, placenta, and uterus in increasing amounts as gestation progresses, in a manner similar to that seen in humans (Moussard et al., 1986; Schellenberg and Kirkby, 1997). Intrauterine prostaglandins promote the parturition of guinea pigs by a mechanism that does not involve luteolysis and systemic progesterone withdrawal, which are essential actions of labor-promoting prostaglandins in murid rodents (Welsh et al., 2005). Therefore, hormonal control of gestation length and parturition is similar in humans and guinea pigs, but quite different from that of rats and mice.

In summary, humans share many more similarities with guinea pigs than rats in regards to pregnancy, gestational synthesis of estrogen and progesterone, intrauterine fetal development, and parturition. The species differences between murid rodents on the one hand, and guinea pigs and humans on the other, are most evident during the second half of gestation and initiation of parturition. This matches the time period in which epoxiconazole-induced adverse effects, namely late resorptions, prolonged gestation length, and dystocia occurred in rats but not in guinea pigs, although the substance was administered at maternally toxic dose levels. In light of these findings, as well as the remarkable species differences between rats and humans, it is therefore very unlikely that the results in rats have any predictive value for human hazard assessment. The results of the present study strengthen the evidence that the guinea pig should be the model of first choice for human risk assessment of chemicals affecting steroid hormone regulation of pregnancy.

CONCLUSION

In conclusion, prenatal or pre- and postnatal administration of epoxiconazole at 50 or 90 mg/kg body weight per day to pregnant guinea pigs caused signs of altered adrenal steroid hormone production, indicating either an alteration in the steroid hormone production of the adrenals or potential consequences of maternal stress.

Mild anemia was observed at the end of gestation. No adverse effects were noted regarding number of implantation sites, pre- and postnatal loss, and number of live and dead fetuses and on fetal sex distribution. Specific adverse, compound-related effects on fetal morphology, in particular well-known triazole-related craniofacial malformations, were not observed. Furthermore, there were no effects on gestation length, parturition, and postnatal development up to and including the dose level of 90 mg/kg body weight per day. In contrast to rats, the histopathology of placentas from guinea pigs treated with epoxiconazole did not reveal any adverse effect.

Distinct species differences in pregnancy, placental function, and parturition, which exist in rats on the one hand and in guinea pigs and humans on the other, probably explain the difference in outcome of the prenatal and pre-postnatal toxicity studies. These differences between rats and guinea pigs are especially evident regarding the hormonal regulation of pregnancy and birth, as well as the site of estrogen and progesterone synthesis. Both humans and guinea pigs switch from luteal to placental progesterone supply in later pregnancy; estrogen is produced mainly by the placenta and the fetus in these species. Estrogen hormonal regulation was especially affected by epoxiconazole treatment in pregnant rats, as shown by markedly reduced serum estrogen levels at the end of gestation. These decreased estradiol concentrations were linked to the increased incidence of late resorptions in rats, which were prevented by maternal estrogen supplementation. In contrast, the serum estrogen concentrations were not altered in guinea pigs. As a result, the incidence of resorptions, gestation length, and parturition were not affected by epoxiconazole administration in this species, in which hormonal regulation during pregnancy is similar to humans, but differs from that of rats. It is therefore very unlikely that the effects on pregnancy and parturition observed in rats can be extrapolated to humans. In contrast, the pregnant guinea pig shares many similarities to pregnant humans regarding hormonal regulation; therefore, the guinea pig is considered to be a suitable experimental animal species to assess potential effects on steroid hormone regulation during human pregnancy.

CONFLICT OF INTEREST

Drs. Schneider, Hofmann, Stinchcombe, Rey Moreno, Fegert, Strauss, Gröters, Fabian, Thiaener, and Fussell, as well as Prof. Dr. van Ravenzwaay are all employees of BASF SE, which manufactures and distributes epoxiconazole.

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